

The Model-Space Closed-Range Dichotomy

A sampling/interpolation theorem prompted by arXiv:2505.15079

Agent lane 2 (agent_lane_02)

Model: GPT5.6

26 August 2026

Status: substantial partial result, likely valid. The packet gives a complete structural dichotomy for all $1 \leq p < \infty$, an exact frame/Riesz criterion for K_θ^2 , and a concrete infinite product-inner classification. It does not geometrically characterize all sampling and interpolation measures for arbitrary θ and p .

1 Source proposal

Dyakonov’s paper [1] classifies closed-range Carleson embeddings for Hardy and Bergman spaces. On page 6 it proposes the analogous study for the star-invariant spaces

$$K_\theta^p := H^p \cap \theta \bar{z} \overline{H^p}, \quad 1 \leq p < \infty,$$

where θ is inner, and predicts that the answer should resemble the Bergman sampling/interpolation pattern rather than the Hardy pattern.

We remark, by way of digression, that a similar study of Carleson measures with the closed range property could definitely be performed for many other function spaces. In particular, various subspaces of H^p and A^p are worth looking at. To mention one specific case of interest—and a possible topic of future research—consider the *star-invariant* (or *model*) *subspace* $K_\theta^p := H^p \cap \theta \bar{z} \overline{H^p}$, with $1 \leq p < \infty$, associated with an inner function θ . In general, Carleson measures for K_θ^p are not completely well understood, and the problem of describing them (first posed in [6]) has a long history. Among the many papers that treat it, we cite [2, 10, 31] and [18, pp. 80–81]. It is conceivable, though, that if the closed range condition is imposed, a neater characterization might come out. Here, we limit ourselves to observing that, while defined in the H^p context, star-invariant subspaces tend to exhibit certain Bergman-type features. For instance, they possess nontrivial sampling measures (see, e.g., [9, Theorem 3] for a class of examples and [3] for further details), their zero-sets vary with p (see [11]), etc. Thus, with regard to the problem at hand, one would expect the situation with K_θ^p to follow the A^p pattern rather than that of H^p .

Figure 1: The complete model-space future-work passage on page 6 of the source PDF.

Throughout, μ is a positive Borel measure on the open disk such that the restriction operator

$$J_\mu : K_\theta^p \longrightarrow L^p(\mu), \quad J_\mu f = f|_{\text{supp } \mu},$$

is bounded. We call μ *sampling* if

$$\|f\|_{H^p} \leq C \|f\|_{L^p(\mu)} \quad (f \in K_\theta^p). \quad (1)$$

If $\mu = \sum_n a_n \delta_{z_n}$, we call the support μ -*interpolating* when J_μ is onto $L^p(\mu)$.

2 Proof intuition

There are only two possible geometries. If restriction is injective, closed range is exactly the reverse estimate (1). If it is not injective, choose a nonzero F in the kernel. The measure is forced onto the discrete zero set of F . More importantly, model spaces allow inner division: removing the full Blaschke factor at one zero keeps the quotient in K_θ^p . Doing this at each support point produces every coordinate vector in the trace range. The range is therefore dense, so closed range is equivalent to interpolation (surjectivity).

In the Hilbert case, restriction is an analysis operator for weighted reproducing kernels. Surjectivity of the analysis operator is equivalent to its synthesis adjoint being bounded below, which is precisely the Riesz-sequence condition. Thus the proposed Bergman-type split becomes a frame-versus-Riesz dichotomy.

3 The structural dichotomy for every finite exponent

Lemma 1 (Star-invariant division). *Let $1 \leq p < \infty$, let $a \in \mathbb{D}$, and let b_a be a Blaschke factor with zero a . If $F \in K_\theta^p$ has a zero of order at least m at a , then $F/b_a^m \in K_\theta^p$.*

Proof. The quotient $G := F/b_a^m$ is analytic and belongs to H^p , since b_a is inner. In terms of boundary functions, write $F = \theta \bar{z} \bar{h}$ with $h \in H^p$. Since $1/b_a^m = \bar{b}_a^m$ almost everywhere on $\mathbb{T}$,

$$G = \theta \bar{z} \overline{b_a^m h}.$$

Thus G belongs to both defining factors of K_θ^p . □

Theorem 1 (Sampling/interpolation dichotomy). *Let θ be a nonconstant inner function, $1 \leq p < \infty$, and suppose $J_\mu : K_\theta^p \rightarrow L^p(\mu)$ is bounded. Its range is closed if and only if exactly one of the following alternatives holds.*

- (S) J_μ is injective and μ is sampling for K_θ^p .
- (I) J_μ is not injective; the measure has the form

$$\mu = \sum_n a_n \delta_{z_n}$$

on a finite or countable discrete zero set, and that support is μ -interpolating for K_θ^p .

In alternative (I), any closed range is automatically all of $L^p(\mu)$.

Proof. Suppose first that J_μ is injective. If its range is closed, then it maps the Banach space K_θ^p isomorphically onto its range. The bounded inverse theorem gives (1). Conversely, that inequality makes J_μ bounded below, hence gives closed range. This proves (S).

Now suppose that the kernel contains a nonzero function F . Since $F = 0$ μ -almost everywhere, continuity forces

$$\text{supp } \mu \subset \{z \in \mathbb{D} : F(z) = 0\}. \tag{2}$$

Indeed, if $F(w) \neq 0$ at a support point, then $|F|$ is bounded away from zero on some neighborhood of w having positive μ -measure, a contradiction. The zero set in (2) is discrete, so $\text{supp } \mu = \{z_n\}$ is countable and $\mu = \sum_n a_n \delta_{z_n}$ with $a_n > 0$.

Let m_n be the exact zero multiplicity of F at z_n . By Lemma 1,

$$G_n := \frac{F}{b_{z_n}^{m_n}} \in K_\theta^p.$$

The function G_n is nonzero at z_n and vanishes at every other support point. After scalar normalization, $J_\mu G_n$ is the n th coordinate vector. Therefore the range of J_μ contains all finitely supported sequences, a dense subspace of $L^p(\mu)$. Its range is closed exactly when it equals $L^p(\mu)$, which is alternative (I). The converse is immediate. □

4 Complete Hilbert-space criterion

For $K_\theta^2 = H^2 \ominus \theta H^2$, write

$$k_z^\theta(w) = \frac{1 - \overline{\theta(z)}\theta(w)}{1 - \bar{z}w}$$

for the reproducing kernel at z .

Theorem 2 (Frame/Riesz form). *Let $J_\mu : K_\theta^2 \rightarrow L^2(\mu)$ be bounded.*

1. *If J_μ is injective, then it has closed range exactly when μ is sampling. For an atomic uniqueness support $\mu = \sum_n a_n \delta_{z_n}$, this is equivalent to $\{\sqrt{a_n} k_{z_n}^\theta\}$ being a frame for K_θ^2 .*
2. *If J_μ is not injective, then $\mu = \sum_n a_n \delta_{z_n}$ as in Theorem 1, and J_μ has closed range exactly when*

$$\{\sqrt{a_n} k_{z_n}^\theta\}_n \tag{3}$$

is a Riesz sequence in K_θ^2 .

Thus these two alternatives classify every bounded closed-range embedding of K_θ^2 into an interior L^2 space.

Proof. Only the noninjective assertion needs proof beyond Theorem 1. Identify $L^2(\mu)$ unitarily with ℓ^2 by $g \mapsto (\sqrt{a_n} g(z_n))_n$. Under this identification, J_μ becomes the analysis operator

$$Tf = (\langle f, \sqrt{a_n} k_{z_n}^\theta \rangle)_n,$$

and

$$T^*(c_n) = \sum_n c_n \sqrt{a_n} k_{z_n}^\theta.$$

Boundedness of T supplies the upper Riesz bound. By Theorem 1, closed range in the noninjective branch is equivalent to surjectivity of T . A Hilbert-space operator is onto exactly when its adjoint is bounded below, which supplies the lower Riesz bound for (3). The atomic injective statement is the usual analysis-operator characterization of a frame. $\square$

5 An exact infinite product-inner family

Corollary 1. *Let B_1 be an infinite interpolating Blaschke product with zeros $\mathcal{Z} = \{z_n\}$, let B_2 be a nonconstant inner function, and put $\theta = B_1 B_2$. For positive weights a_n , define $\mu = \sum_n a_n \delta_{z_n}$. Then*

$$J_\mu : K_{B_1 B_2}^2 \longrightarrow L^2(\mu)$$

is bounded if and only if

$$\sup_n \frac{a_n}{1 - |z_n|^2} < \infty. \tag{4}$$

Under this boundedness condition, its range is closed if and only if

$$0 < \inf_n \frac{a_n}{1 - |z_n|^2} \leq \sup_n \frac{a_n}{1 - |z_n|^2} < \infty. \tag{5}$$

In the closed-range case the map is onto and

$$\ker J_\mu = B_1 K_{B_2}^2.$$

Proof. The standard orthogonal decomposition

$$K_{B_1 B_2}^2 = K_{B_1}^2 \oplus B_1 K_{B_2}^2$$

shows that restriction to $\mathcal{Z}$ kills the second summand. Because B_1 is interpolating, the normalized kernels

$$\frac{\sqrt{1 - |z_n|^2}}{1 - \bar{z}_n z}$$

form a Riesz basis for $K_{B_1}^2$. At a zero z_n of B_1 , the $K_{B_1 B_2}^2$ kernel is the same Szegő kernel. Consequently the weighted family in (3) is Bessel exactly under (4), and is Riesz exactly under (5). Theorem 2 gives the range assertions. The orthogonal decomposition also makes the displayed kernel identity exact. $\square$

6 Verification, novelty, and remaining scope

The accompanying verifier checks 500 deterministic finite product-inner models using a Malmquist–Walsh basis. It verifies full row rank, the expected $\deg B_2$ nullity, and agreement between the restriction and kernel Gram matrices. These computations are sanity checks, not the proof.

The run indexes and bounded exact web/arXiv searches on 26 August 2026 used the source id and combinations of “closed range”, “model space”, “star-invariant”, “sampling”, “interpolation”, and “Riesz sequence”. They found the source, the reverse-Carleson theory for model spaces [2], and interpolation work, but no statement of the noninjective density mechanism or Theorems 1–2. This is bounded novelty evidence, not an exhaustive priority claim.

The broad source program is not finished: geometric tests for all sampling and interpolation measures remain difficult and depend on θ and p . The packet settles the operator-theoretic architecture, completely settles the Hilbert criterion in frame/Riesz terms, and gives a nontrivial infinite family with an explicit weight characterization.

7 Human-review recommendation

Review as a substantial partial result. The crucial audit points are Lemma 1, the support implication (2), and the closed-range-to-surjectivity step. The frame/Riesz theorem then follows from standard Hilbert closed-range duality.

References

- [1] K. M. Dyakonov, *Carleson-type embeddings with closed range*, arXiv:2505.15079, 2025–2026.
- [2] A. Blandignères, E. Fricain, F. Gaunard, A. Hartmann, and W. T. Ross, *Reverse Carleson embeddings for model spaces*, arXiv:1205.3260, 2012.
- [3] E. Fricain and J. Mashregi, *Introduction to Model Spaces and Their Operators*, Cambridge University Press, 2016.